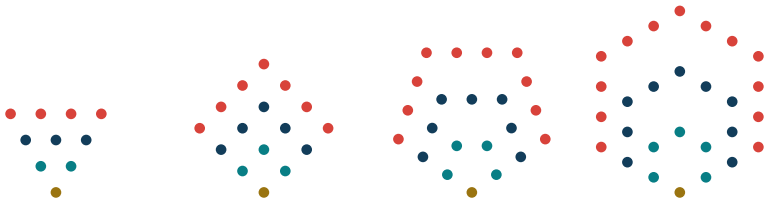
NUMBER PATTERNS ON ONE PAGE

Polygonal numbers

Dots on nested polygons give triangular, square and pentagonal numbers; negative indices lead to Euler.

1 / THE RULE AND THE PICTURE

The nth s-gonal number is the number of dots in the first n nested s-gons



triangular $1, 3, 6, 10$ — square $1, 4, 9, 16$ — pentagonal $1, 5, 12, 22$ — hexagonal $1, 6, 15, 28$

s-gons of side $0, 1, 2, 3$ steps share one corner; each colour is a new layer

Place regular s-gons with sides of $0, 1, 2, \ldots$ steps in one corner and put a dot at every step along the sides. The nth polygon adds a layer of $1 + (n-1)(s-2)$ new dots, since the two sides at the shared corner are already there, so the layers grow by $s-2$ each time and

$$P_s(n) = \sum_{j=1}^{n} \big[1 + (j-1)(s-2)\big]$$

which simplifies to

$$\boxed{P_s(n) = \tfrac{(s-2)n^2 - (s-4)n}{2}}$$

2 / THE FIRST POLYGONAL NUMBERS

Each row has a quadratic formula, and its second difference is $s-2$

s	name	$P_s(n)$	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
3	triangular	$\frac{n(n+1)}{2}$	1	3	6	10	15	21	28	36
4	square	n^2	1	4	9	16	25	36	49	64
5	pentagonal	$\frac{n(3n-1)}{2}$	1	5	12	22	35	51	70	92
6	hexagonal	$n(2n-1)$	1	6	15	28	45	66	91	120
7	heptagonal	$\frac{n(5n-3)}{2}$	1	7	18	34	55	81	112	148
8	octagonal	$n(3n-2)$	1	8	21	40	65	96	133	176

Taking differences of any row gives $1 + (n-1)(s-2)$, then the constant $s-2$, which is why every row is quadratic in n.

3 / THREE RELATIONS

Triangles build every polygon, and a square test recognises the numbers

Cutting an s-gon from one corner gives $s-2$ triangles, so $P_s(n) = n + (s-2)T_{n-1}$ with $T_m = \frac{m(m+1)}{2}$; for $s=5$, $n=4$ this is $22 = 4 + 3\cdot 6$. Going from $s-1$ sides to s sides adds a triangular number, $P_s(n) = P_{s-1}(n) + T_{n-1}$, so $22 = 16 + 6$. Completing the square in $P_s(n)$ gives

$$\boxed{8(s-2)\,N + (s-4)^2 = \big(2(s-2)n - (s-4)\big)^2}$$

so a number N is s-gonal when the left side is a perfect square of this form. For $s=3$ it reads $8N+1 = (2n+1)^2$, for $s=4$ it is $16N = (4n)^2$, and for $s=5$ it is $24N+1 = (6n-1)^2$.

4 / NEGATIVE INDICES

The same formula at $n = 0, -1, -2, \ldots$ gives further numbers

s	formula at $-n$	$n = 1, \ldots, 6$	what appears
3	$P_3(-n) = \frac{n(n-1)}{2}$	$0, 1, 3, 6, 10, 15$	the triangular numbers again
4	$P_4(-n) = n^2$	$1, 4, 9, 16, 25, 36$	the squares again
5	$P_5(-n) = \frac{n(3n+1)}{2}$	$2, 7, 15, 26, 40, 57$	new numbers
6	$P_6(-n) = n(2n+1)$	$3, 10, 21, 36, 55, 78$	new, and triangular: T_{2n}

For $s=3$ and $s=4$ the negative indices repeat numbers already found, but from $s=5$ they give new ones. Together with $P_5(n)$ they form the generalised pentagonal numbers $0, 1, 2, 5, 7, 12, 15, 22, 26, 35, 40, \ldots$, where $24N+1 = (6n-1)^2$ holds for every integer n, positive or negative. These are the exponents that survive when $(1-x)(1-x^2)(1-x^3)\cdots$ is multiplied out.

5 / NUMBERS THAT BELONG TO TWO FAMILIES

Coincidences between the rows

Every hexagonal number is triangular, because $P_6(n) = n(2n-1) = T_{2n-1}$, so the hexagonal numbers are the triangular numbers of odd index. Numbers that are both triangular and square are $1, 36, 1225, 41616, 1413721, \ldots$, since $\frac{m(m+1)}{2} = k^2$ is the Pell equation $(2m+1)^2 - 8k^2 = 1$. Numbers that are both triangular and pentagonal begin $1, 210, 40755, 7906276$, and the first number after 1 that is pentagonal and hexagonal is $40755 = T_{285} = P_5(165) = P_6(143)$.

ASIDE / THREE PRODUCTS, THREE SHAPES

triangular: $\prod (1-x^n)^3 = \sum_{n\geq 0} (-1)^n (2n+1) x^{n(n+1)/2}$

square: $\prod \frac{1-x^n}{1+x^n} = \sum_{n\in\mathbb{Z}} (-1)^n x^{n^2}$

pentagonal: $\prod (1-x^n) = \sum_{n\in\mathbb{Z}} (-1)^n x^{n(3n-1)/2}$

LINKS TO OTHER SHEETS

Euler's pentagonal formula, which follows, proves the third of these; the partitions sheet counts the same structures.

Convention. $P_s(n)$ the nth s-gonal number; $T_m = m(m+1)/2$ the triangular numbers; the three product identities were checked as power series to x^{40}.

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